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Class: 4 MCA B

Lab 1

Aim: Create and configure an AWS S3 Bucket

Step:

1. Log in to your account to start the Lab
2. Navigate to S3 Services
3. Create Bucket
4. Enter a unique name for the S3 Bucket
5. Bucket Settings (keep it default)
 - a. Object Ownership - default
 - b. Block public access - enabled
 - c. Enable Bucket versioning
 - d. Create Bucket
 - e. Upload, Download files

(S3 Bucket is created)

6. Configure Bucket versioning

Open Bucket → Properties → Bucket Versioning → Click Edit → Enable → Save

7. Configure Lifecycle Rules

Management → Create Rule → Enter Name → Enter prefix → Select Lifecycle rule actions - Expire current versions of objects → Days after object creation - 30 → Save Rule

8. Configure Encryption

Properties → Default Encryption → Save Server-side encryption with Amazon S3 managed keys (SSE-S3)

1. What is S3?

- An S3 bucket is a cloud storage container for storing files such as documents, images, and datasets
- Each file stored in a bucket is called an object, and it can be accessed or managed individually
- Every bucket must have a unique name so it can be identified across all AWS users
- Buckets are created in specific regions, which affects speed, availability, and cost
- You control access permissions to decide who can view, upload, or download files

2. Benefits of S3

- Data performance and durability - Scale storage resources to meet fluctuating demands, with no upfront investments or resource procurement cycles. Your data is available when needed and protected against failures, errors, and threats
- Security, compliance, and auditing - Secure your data against unauthorised access with encryption and access management. S3 also maintains compliance programs and supports numerous auditing capabilities
- Flexible storage options - Save costs without sacrificing performance. Store data across a wide range of cost-effective storage classes that support different data access levels and are designed for specific use cases
- Granular data control - Classify, manage, and report on your data using a variety of storage management features. Log activities, define alerts, and automate workflows without managing additional infrastructure

3. Use cases of S3

- Backup and restore - Build scalable, durable, and secure backup and restore solutions to augment or replace existing on-premises capabilities with S3 and other AWS services, such as Amazon S3 Glacier, Amazon Elastic File System (Amazon EFS), and Amazon Elastic Block Store (Amazon EBS)
- Disaster Recovery - Protect critical data, applications, and IT systems that are running in the AWS Cloud or your on-premises environment without incurring the expense of a second physical site
- Archive - Retire physical infrastructure, and archive data with S3 storage classes that retain objects long-term at the lowest rates
- Data Lakes and Big Data Analytics - Accelerate innovation by creating a data lake in S3. Extract valuable insights using query-in-place, analytics, and machine learning features
- Hybrid cloud storage - Create a seamless connection between on-premises applications and S3 with AWS Storage Gateway. Reduce your data center footprint and use the scale, reliability, and durability of AWS and its innovative machine learning and analytics capabilities
- Cloud-native application data - Build fast, cost-effective mobile and internet-based applications by using AWS services and S3 to store production data

4. Case studies of S3 Bucket

- Georgia-Pacific
- Nasdaq
- Sysco
- Nielsen

5. What is ACL - Access Control List?

- ACLs define who can access your bucket or objects
- They specify what actions (read, write, delete) each user can perform
- They are an older way of managing permissions, now often replaced by bucket policies

6. What is Block Public Access?

- It prevents your bucket or files from being accidentally exposed to everyone online
- It overrides ACLs and policies that might allow public access
- It is a recommended default safety setting for all buckets

7. What is Bucket Versioning?

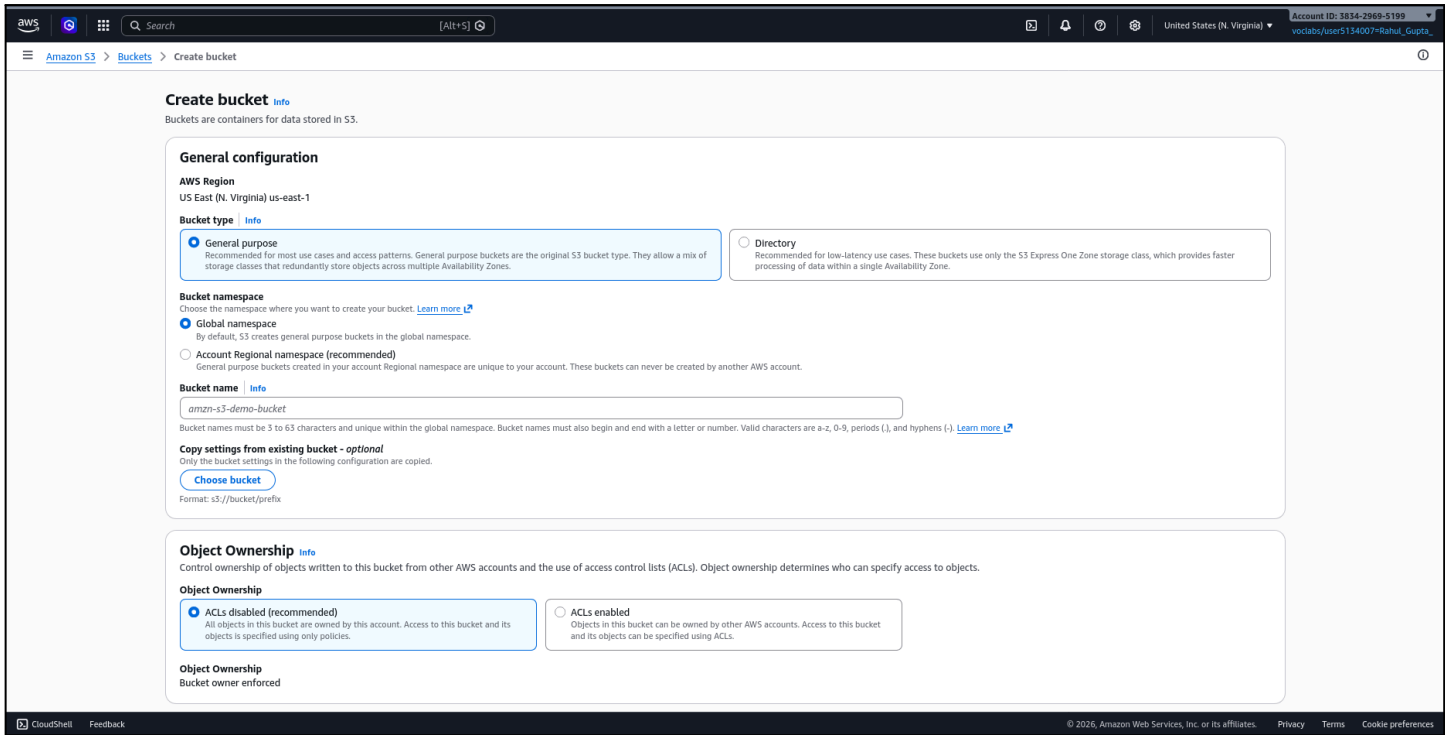
- It keeps multiple versions of the same file in a bucket
- It allows you to recover old versions if files are deleted or overwritten
- It increases storage usage since every version is saved

8. What is an S3 Bucket Lifecycle Rule?

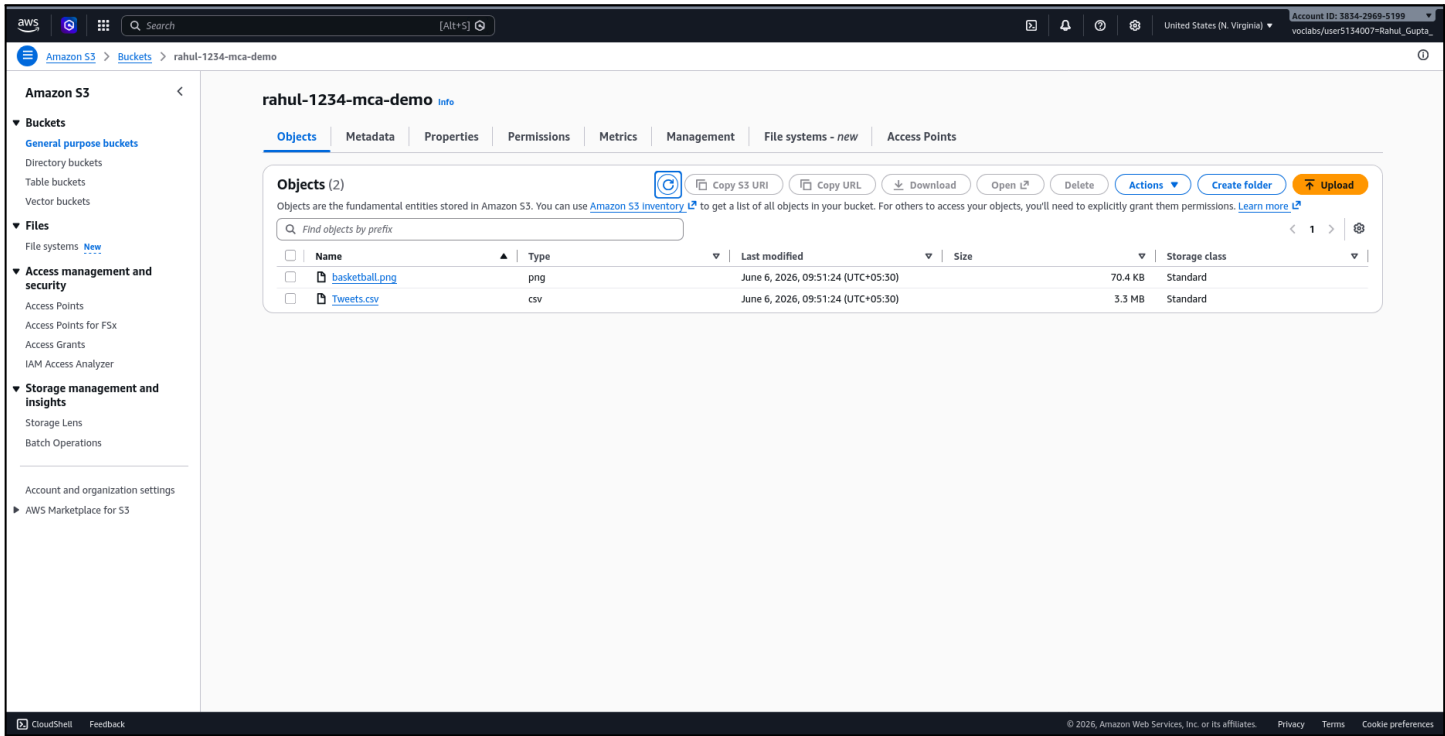
- Lifecycle rules automatically manage files over time
- They can move files to cheaper storage (like Glacier) after a set period
- They can delete old files to save costs and keep storage clean

Output:

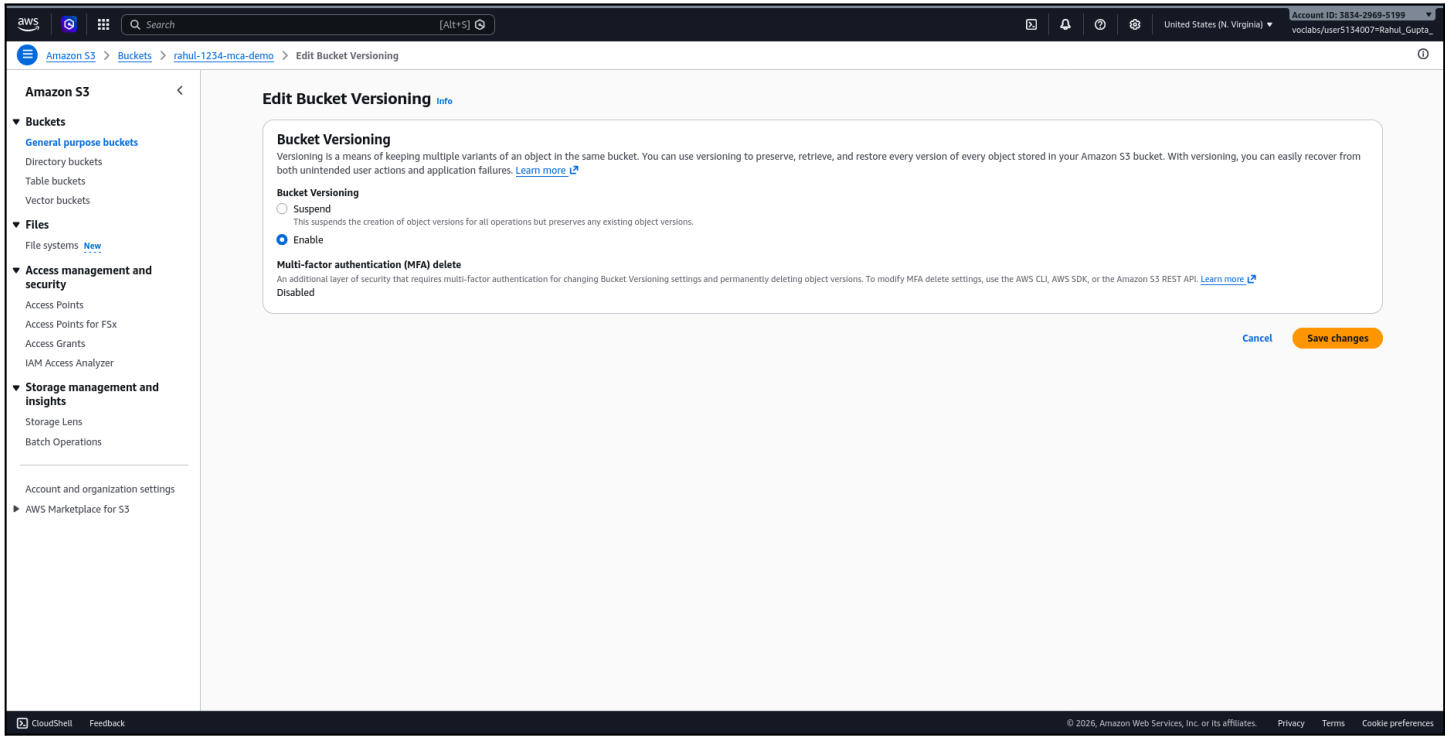
Create Bucket



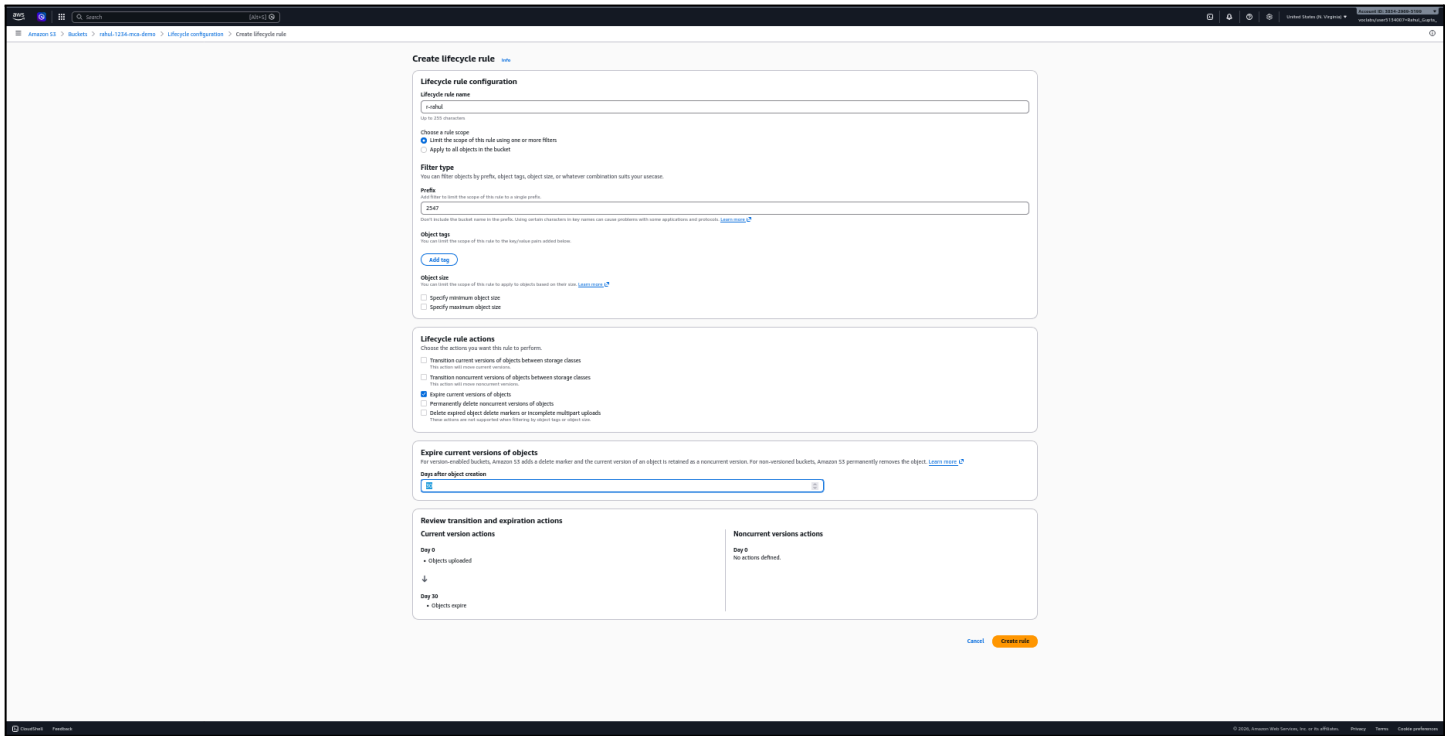
Upload Files in Bucket

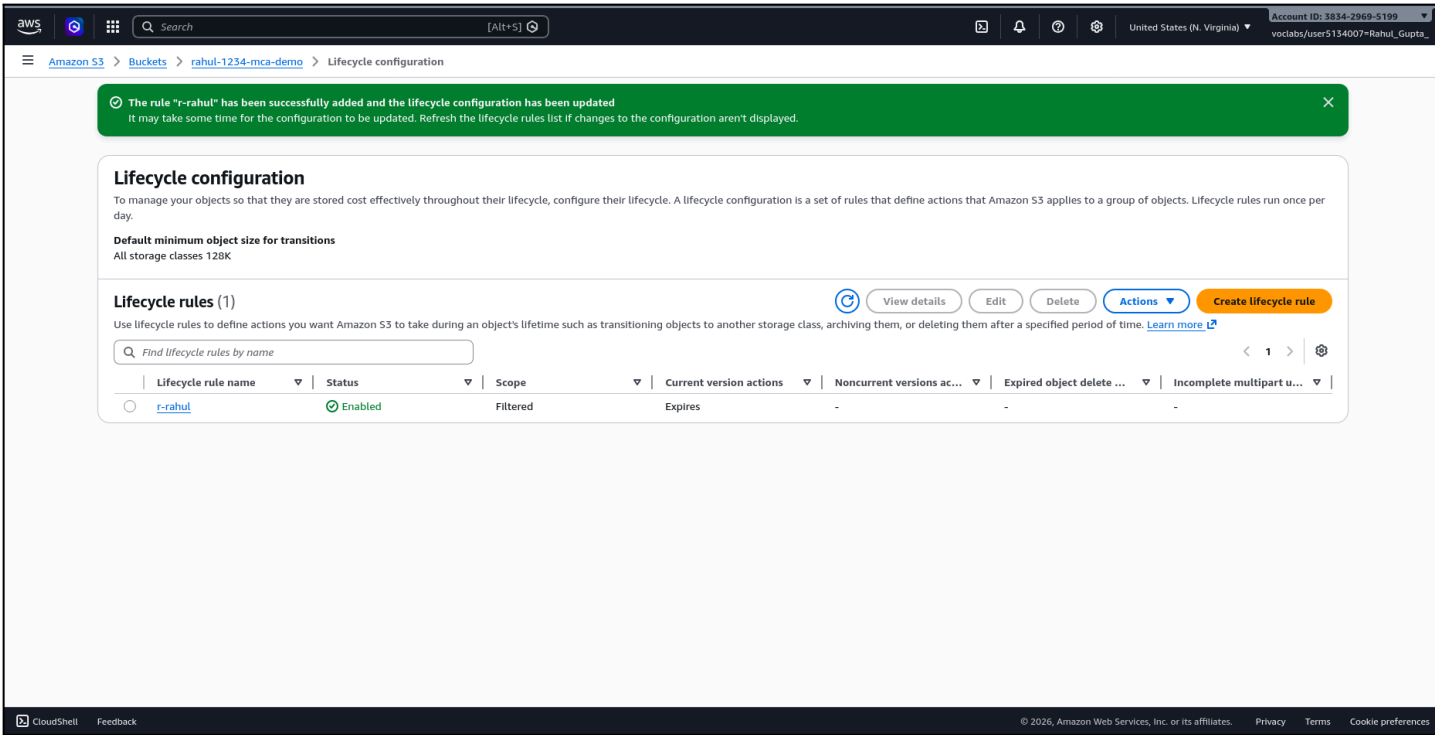


Enable Bucket Versioning

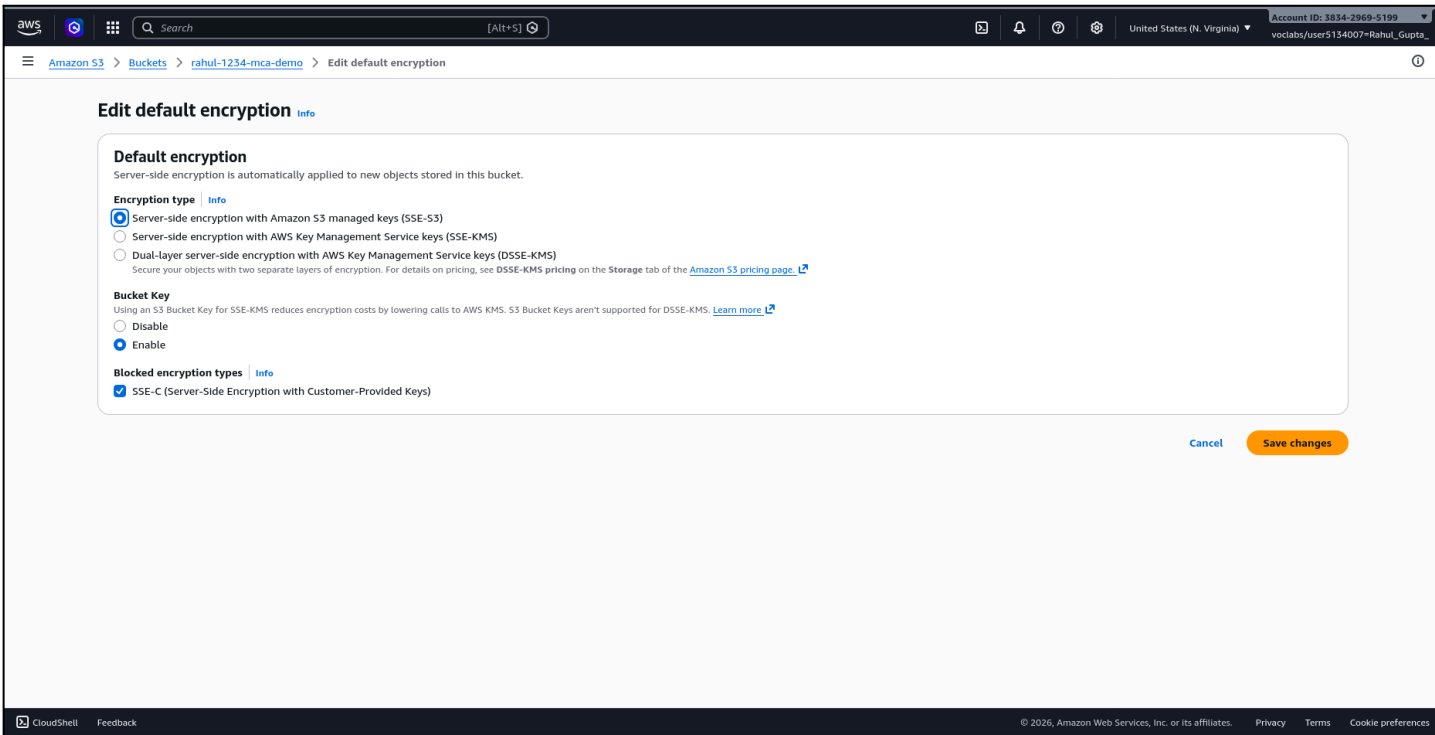


Configure Lifecycle Rules

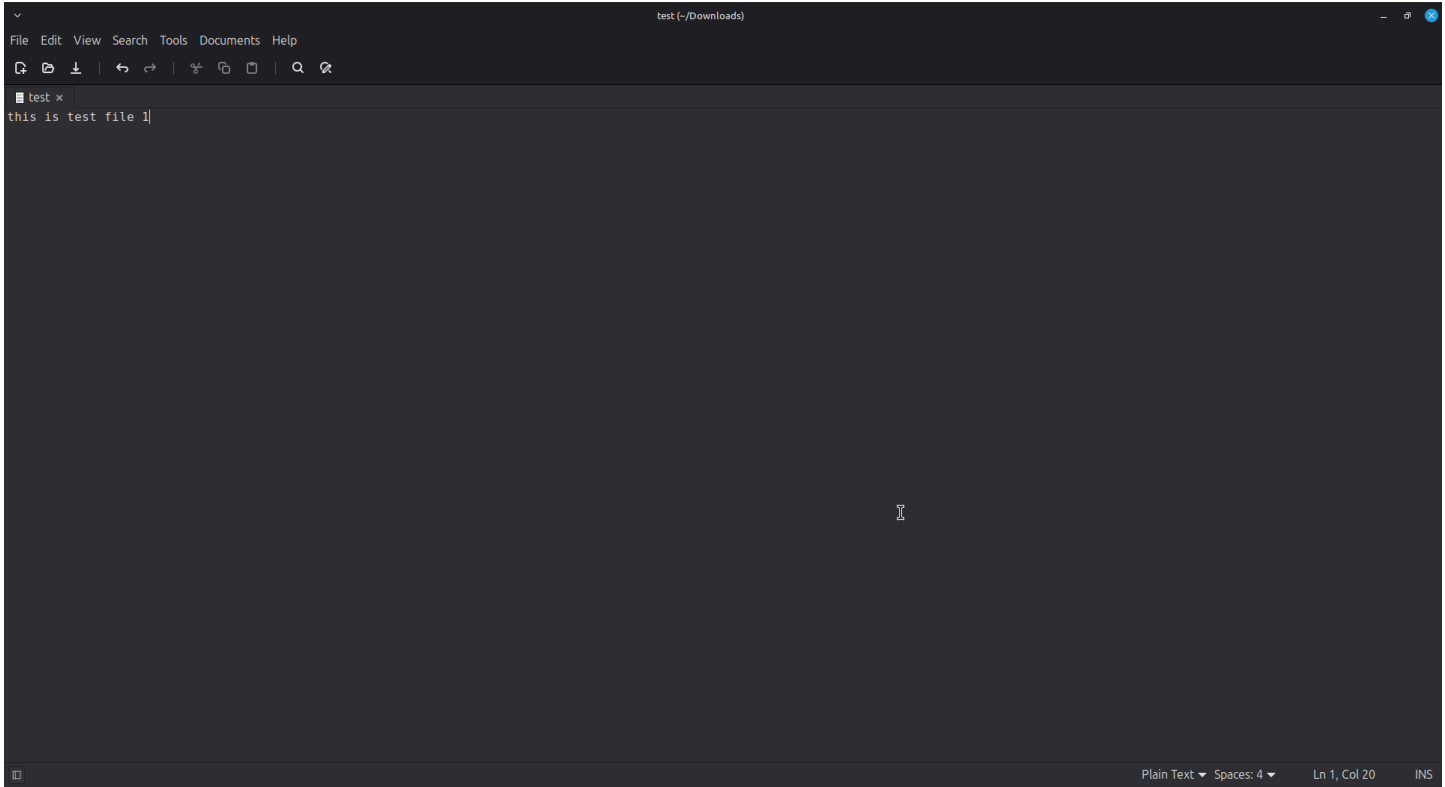
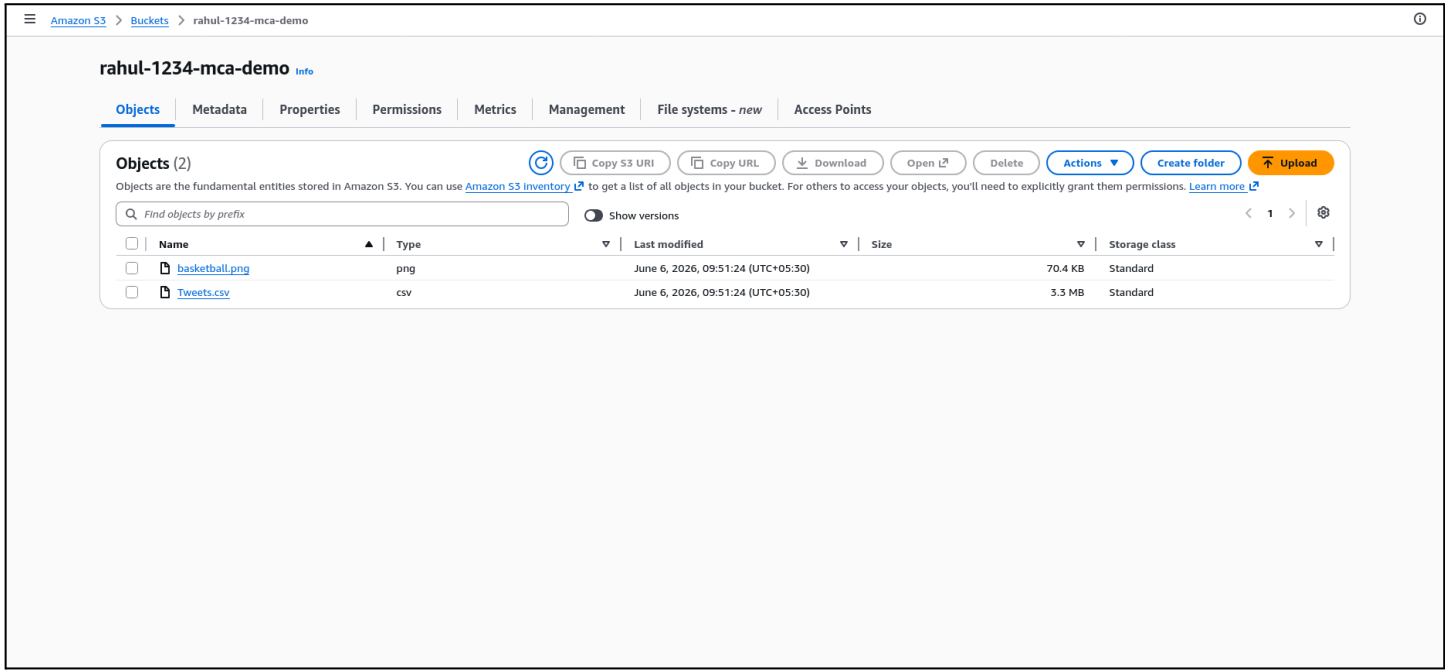




Default Encryption



Bucket Versioning



aws

Search

[Alt+S]

United States (N. Virginia)

Account ID: 3834-2969-5199
voclabs/user5134007-Rahul_Gupta_

Amazon S3

Upload succeeded
For more information, see the [Files and folders](#) table.

Close

Upload: status

After you navigate away from this page, the following information is no longer available.

Summary

Destination
s3://rahul-1234-mca-demo

Succeeded

1 file, 20.0 B (100.00%)

Failed

0 files, 0 B (0%)

Files and folders

Configuration

Files and folders (1 total, 20.0 B)

Find by name

< 1 >

Name	Folder	Type	Size	Status	Error
test	-	-	20.0 B	Succeeded	-

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Search

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Account ID: 3834-2969-5199
voclabs/user5134007-Rahul_Gupta_

Amazon S3 > Buckets > rahul-1234-mca-demo

rahul-1234-mca-demo

Objects

Metadata

Properties

Permissions

Metrics

Management

File systems - new

Access Points

Objects (3)

Copy S3 URI

Copy URL

Download

Open

Delete

Actions

Create folder

Upload

Objects are the fundamental entities stored in Amazon S3. You can use [Amazon S3 Inventory](#) to get a list of all objects in your bucket. For others to access your objects, you'll need to explicitly grant them permissions. [Learn more](#)

Find objects by prefix

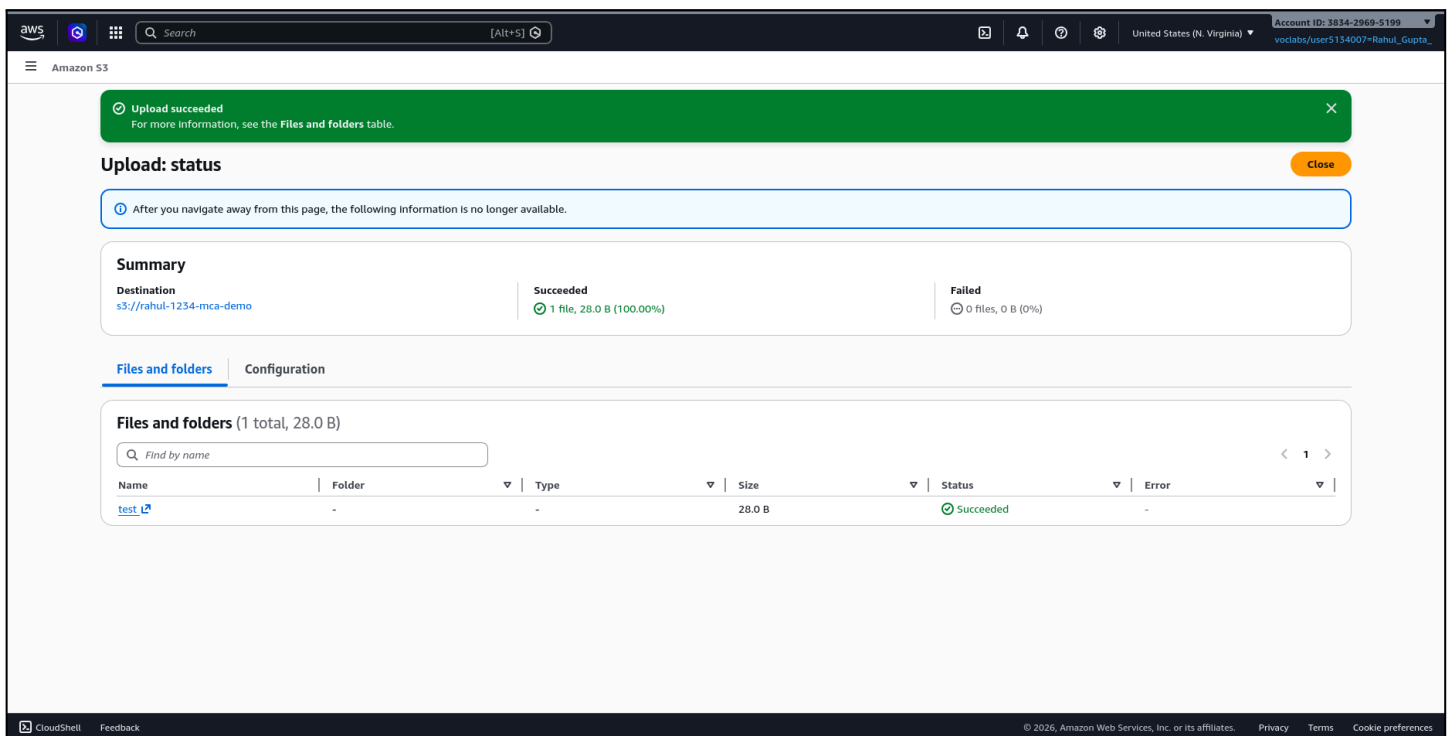
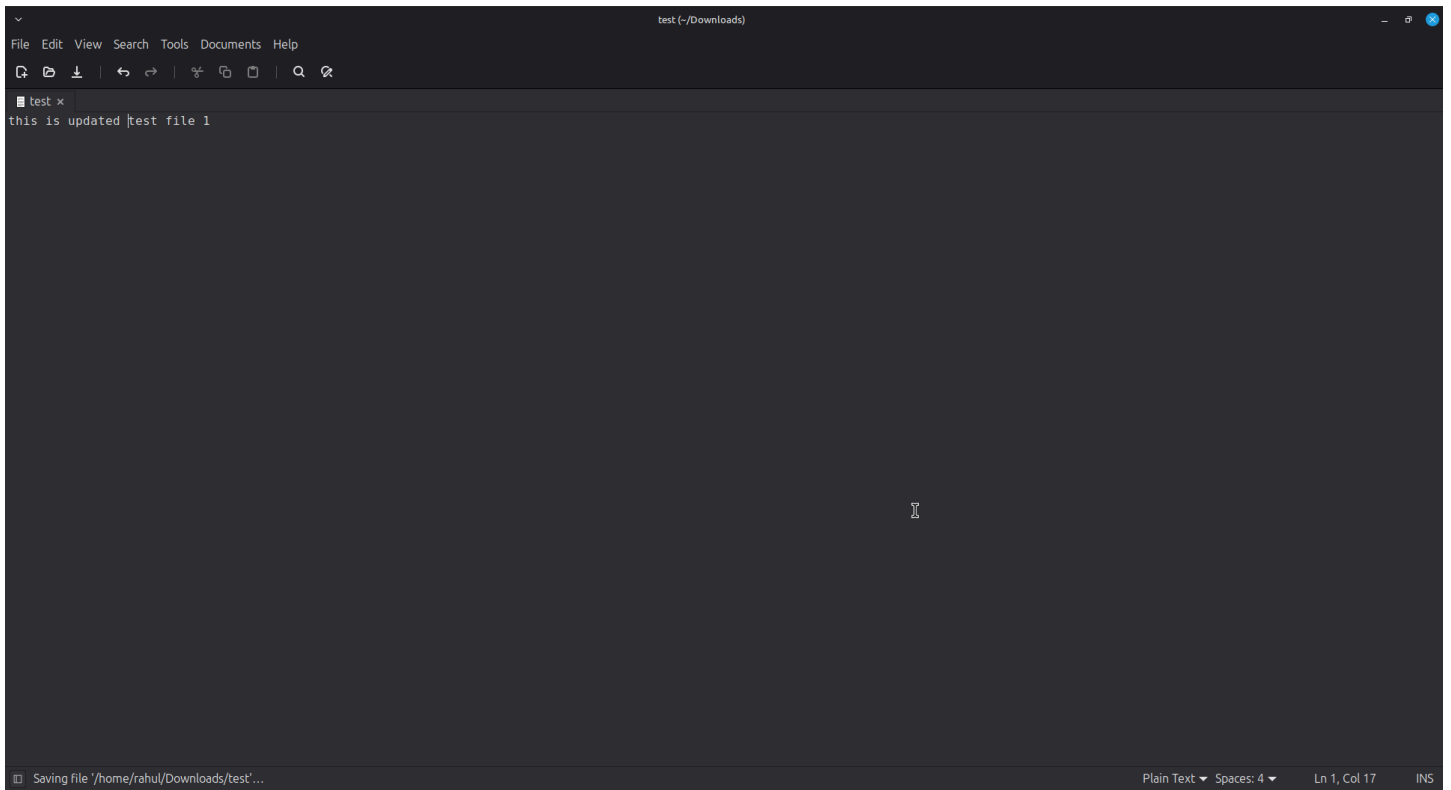
Show versions

< 1 >

☐	Name	Type	Last modified	Size	Storage class
☐	basketball.png	png	June 6, 2026, 09:51:24 (UTC+05:30)	70.4 KB	Standard
☐	test	-	June 8, 2026, 08:29:43 (UTC+05:30)	20.0 B	Standard
☐	Tweets.csv	csv	June 6, 2026, 09:51:24 (UTC+05:30)	3.3 MB	Standard

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Amazon S3

Buckets

rahul-1234-mca-demo

rahul-1234-mca-demo

Info

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Metrics

Management

File systems - new

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Copy URL

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Delete

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Upload

Find objects by prefix

Show versions

< 1 >

☐	Name	Type	Last modified	Size	Storage class
☐	 basketball.png	png	June 6, 2026, 09:51:24 (UTC+05:30)	70.4 KB	Standard
☐	 test	-	June 8, 2026, 08:30:43 (UTC+05:30)	28.0 B	Standard
☐	 Tweets.csv	csv	June 6, 2026, 09:51:24 (UTC+05:30)	3.3 MB	Standard

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Amazon S3

Buckets

rahul-1234-mca-demo

test

test

Info

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Versions

Copy S3 URI

Download

Open

Object actions

Versions (5)

Download

Open

Delete

Actions

< 1 >

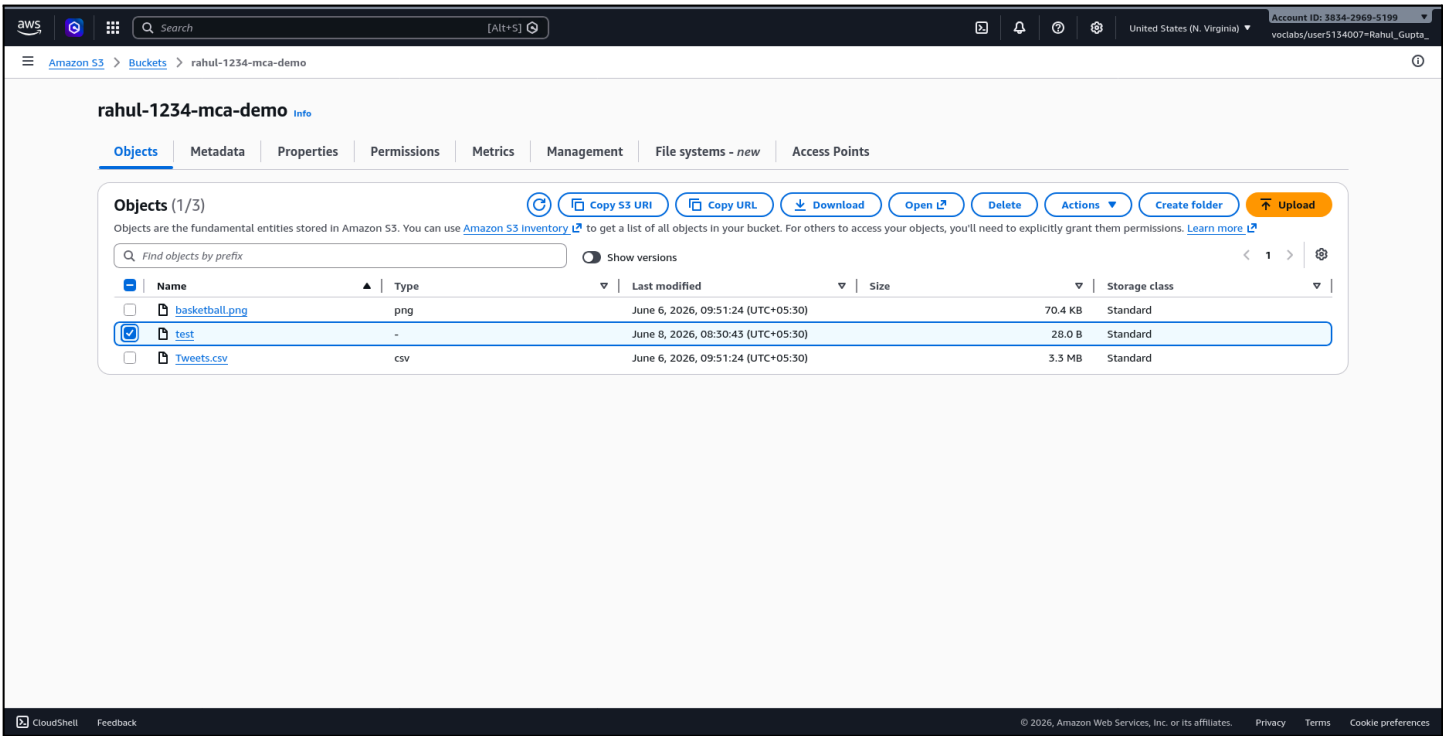
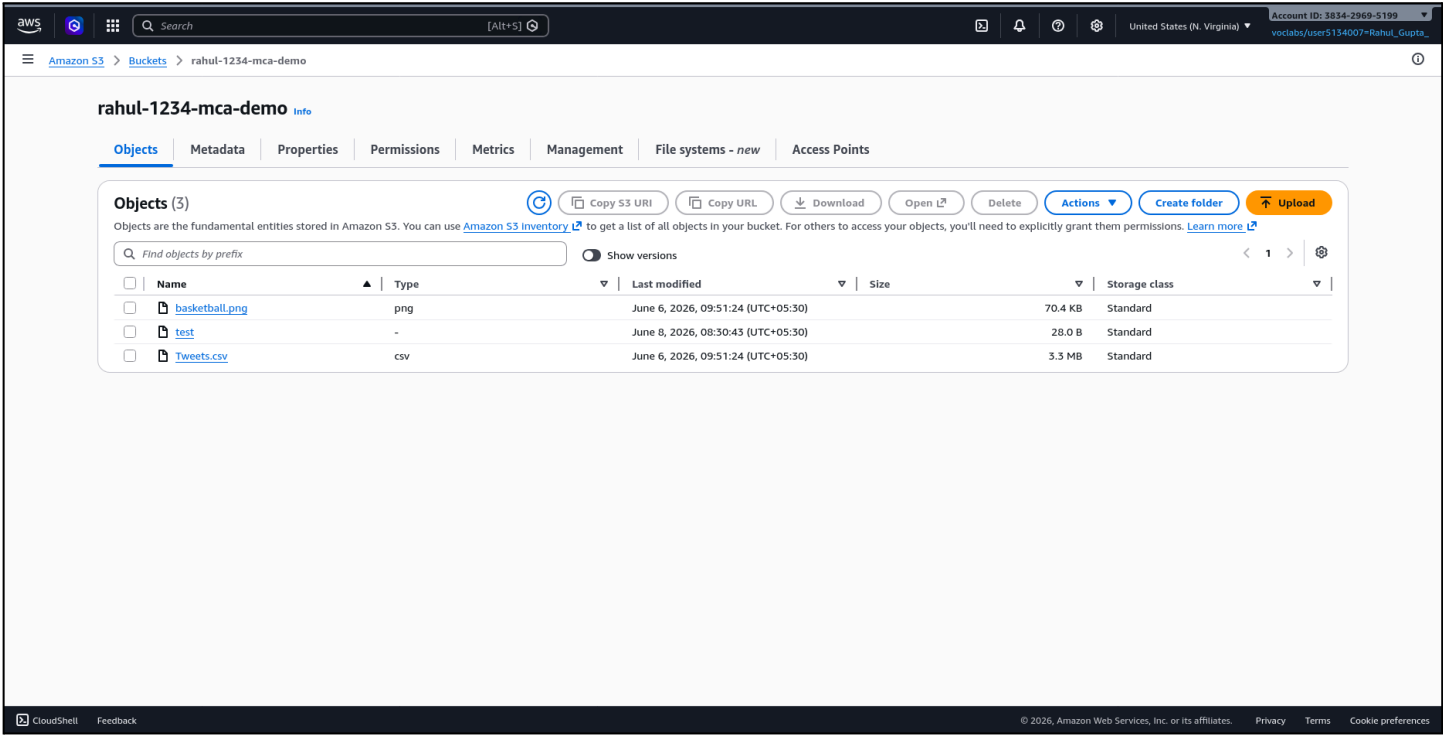
☐	Version ID	Type	Last modified	Size	Storage class
☐	 VHuypkxp5fJxNarRoMRQ2xFJNZ5Ta9... (Current version)	-	June 8, 2026, 08:30:43 (UTC+05:30)	28.0 B	Standard
☐	 S1fStEwTqnlVz5E6.WdLKLI0_2NM...	-	June 8, 2026, 08:29:43 (UTC+05:30)	20.0 B	Standard
☐	 cL_5.VsVL6EWm6By2WuJvUzplQluAPk	Delete marker	June 8, 2026, 08:27:27 (UTC+05:30)	0 B	-
☐	 o0GUmjCwkI47dFylatmqY3aw5.8...	-	June 8, 2026, 08:24:55 (UTC+05:30)	28.0 B	Standard
☐	 U.KTG5Jhr3HwfBB5MZXPv076m1...	-	June 8, 2026, 08:24:18 (UTC+05:30)	20.0 B	Standard

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Deleting Object in S3 Bucket



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voclabs/user5114007=Rahul_Gupta_

United States (N. Virginia)

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Amazon S3

Buckets

rahul-1234-mca-demo

rahul-1234-mca-demo

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Standard